LaTeX Squiggly



The iPhone guide

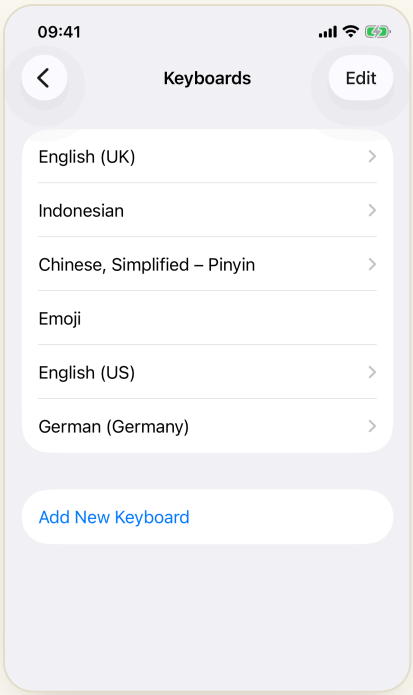
A keyboard for writing maths anywhere. Type `\alpha` and get α. Write an equation and copy it as a picture.

Inside

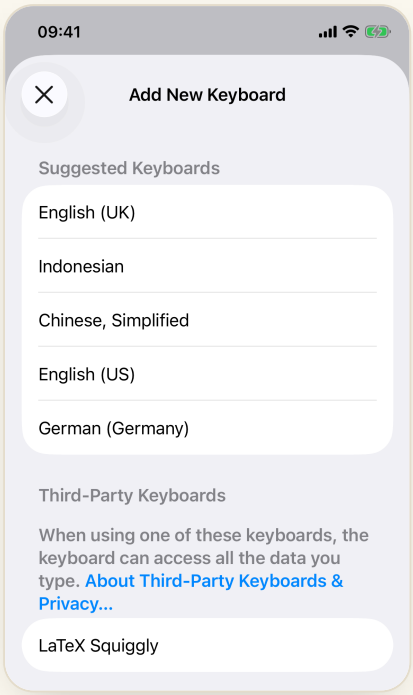
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1 Set up the keyboard

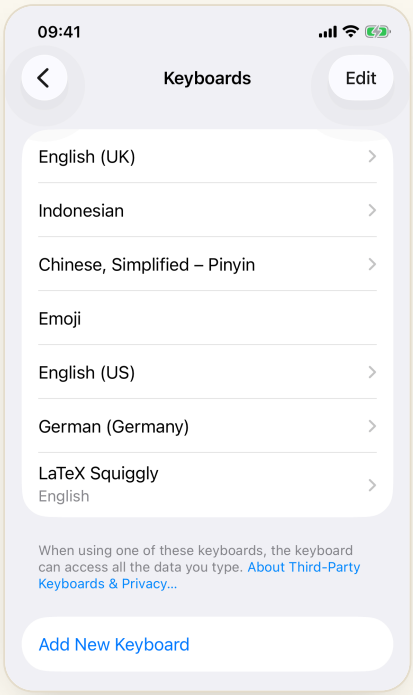
iPhone keyboards are turned on in Settings. You do this once.



1. Open **Settings**, then **General**, **Keyboard**, **Keyboards**. Tap **Add New Keyboard**.



2. Under Third-Party Keyboards, choose **LaTeX Squiggly**.



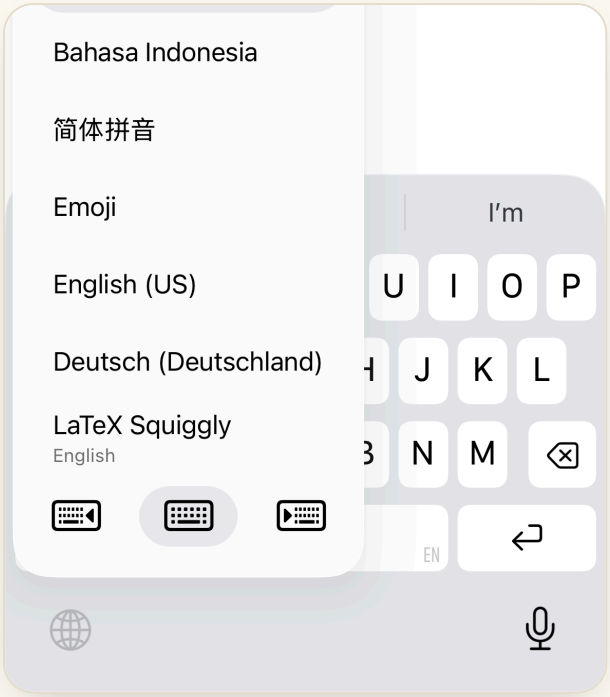
3. Tap **LaTeX Squiggly** and turn on **Allow Full Access**.



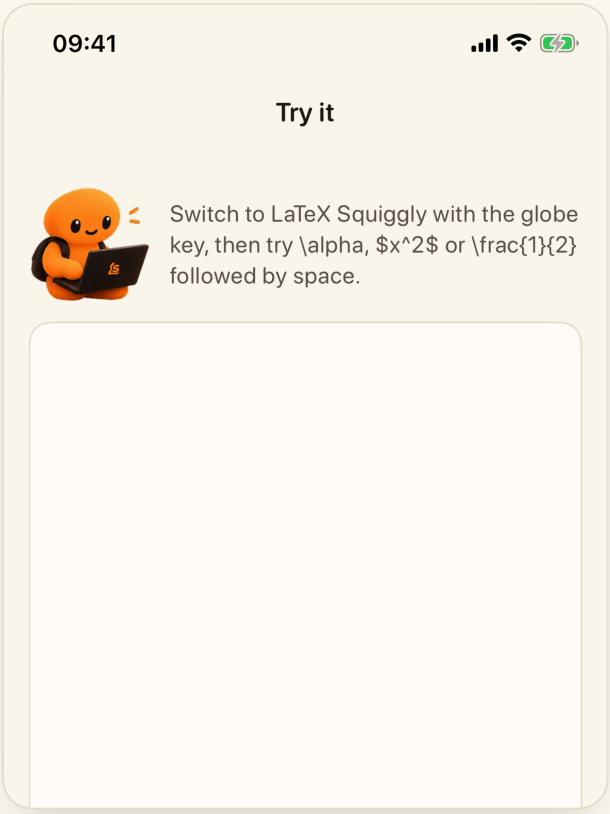
Why Full Access? Only to put equation images on the clipboard, so you can paste them. Everything else works without it. Nothing you type leaves your phone: the keyboard makes no network connections and stores nothing you write.

2 Switch to it

In any app, **touch and hold the globe** 🌐 below the keys, then slide to **LaTeX Squiggly**. A quick tap on the globe moves to the next keyboard.



Hold the globe to pick a keyboard.

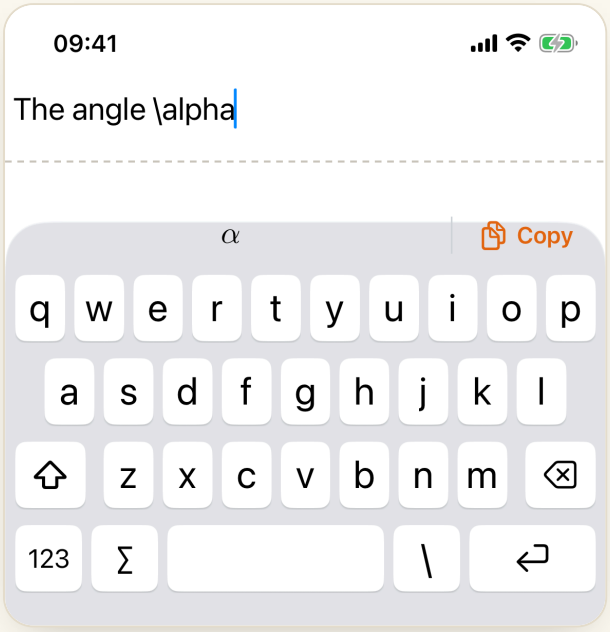


Practise in the app's **Try it** tab. The mascot cheers your first symbol.

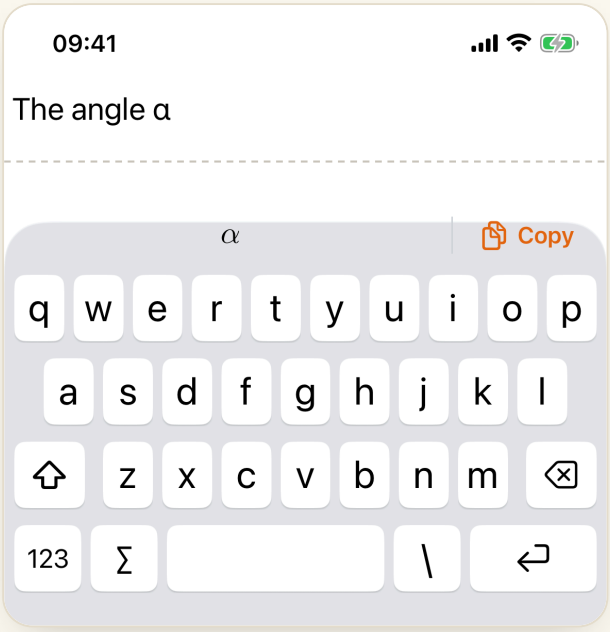
It types like Apple's keyboard: letters pop up as you press them, a touch between two keys goes to the one you most likely meant, and you can slide your finger to the right key before letting go. Hold space and slide to move the cursor. Hold delete and it speeds up to whole words.

3 Type symbols: LaTeX to Unicode

Type a LaTeX command, then **space**. It turns into the real symbol, which anyone can read: in a message, an email, a document.



Type `\alpha`...



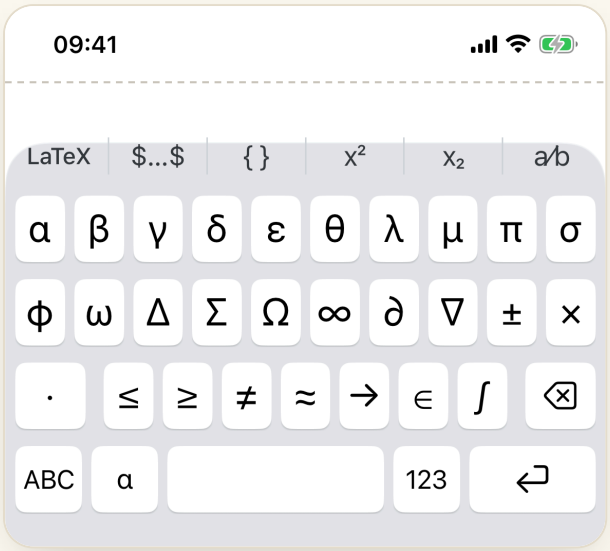
...press space, and it is α.

You type	You get	You type	You get
`\alpha`	α	`\infty`	∞
`\to`	→	`\leq`	≤
`\sum`	∑	`\mathbb{R}`	ℝ
`\frac{1}{2}`	½	`\pm`	±

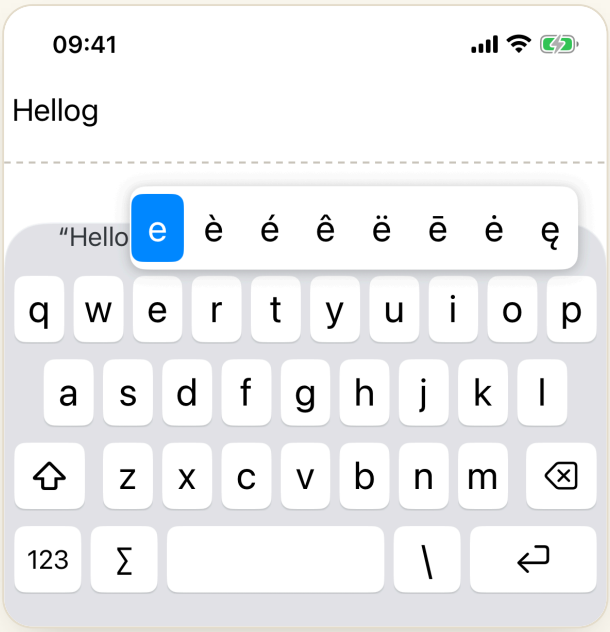
Changed your mind? Press delete straight after a symbol appears and your LaTeX comes back.

4 The ∑ page and held keys

Tap ∑ for Greek letters and maths symbols, one tap each. ABC goes back to the letters.



The ∑ page.



Hold a key for more: accents on letters.

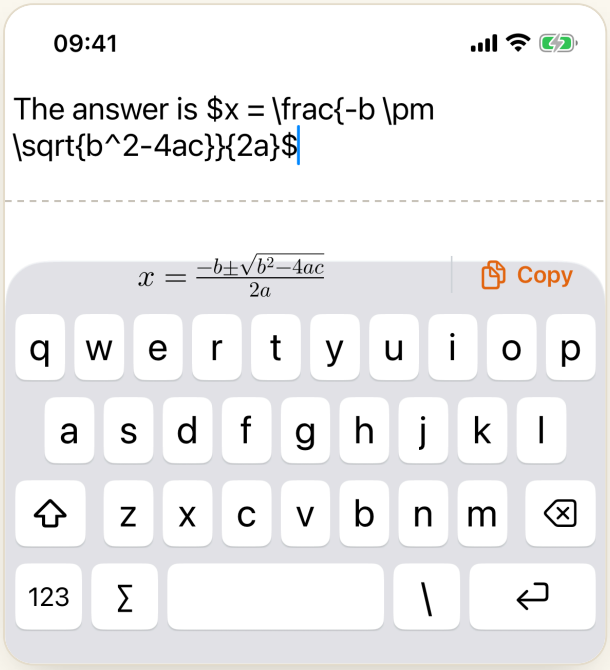
Hold a key to see related characters, then slide to one and let go. Letters offer accents (é, ñ, ü). Maths keys offer their relatives:

Hold	For	Hold	For
=	≠ ≈ ≡ ≅	ε	ϵ
≤	≪ ⊆ ≺	θ	ϑ ⊖
→	← ↔ ⇒ ⇔ ↦	∫	∬ ∮ ∑ ∏

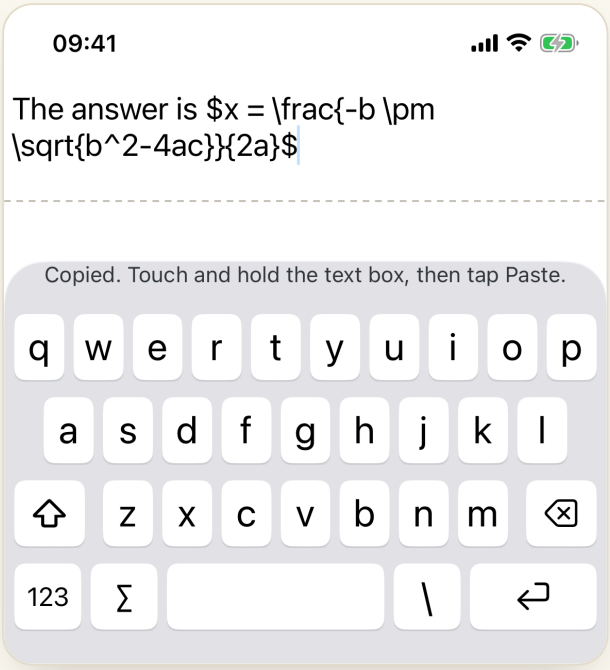
5 Equations as images

Some maths can't be written as plain text: fractions stacked on top of each other, square roots over long expressions, matrices. For these, make a picture.

1. Write your equation **between $ signs**, like `$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$`. Inside the $ signs nothing turns into symbols, so you can take your time.
2. The bar above the keys shows it drawn properly. Tap **Copy**.
3. Touch and hold where you want it, then tap **Paste**.



The bar shows your equation. Tap **Copy**.

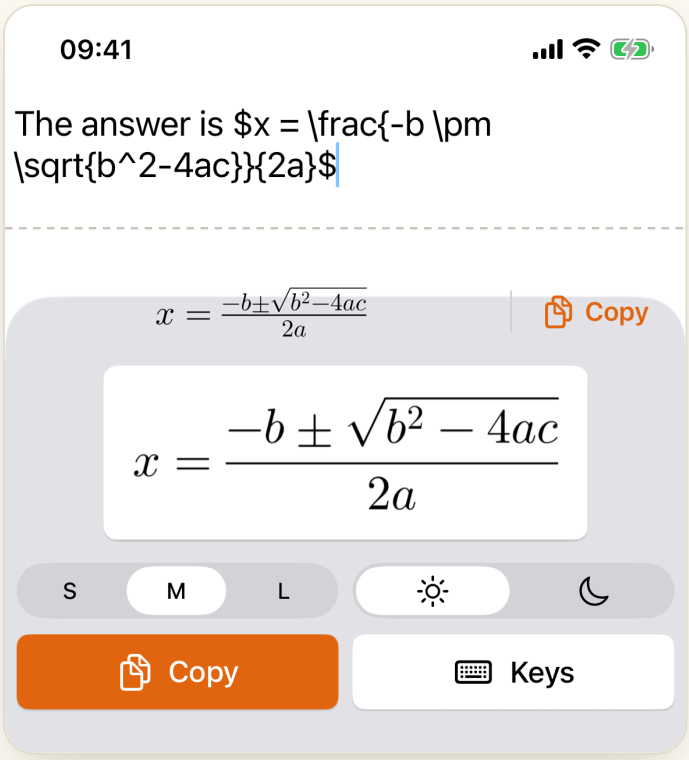


Copied. Now paste it into your message.

> Tap **$...$** in the bar to get both $ signs with the cursor between them.

6 The big preview

Tap the equation in the bar to see it full size, in place of the keys.



The preview, with its options.

S, M, L choose how big the picture is.

The **sun** and **moon** buttons choose a white or a dark background, to suit a light or dark chat.

Copy copies it, just like the bar's Copy.

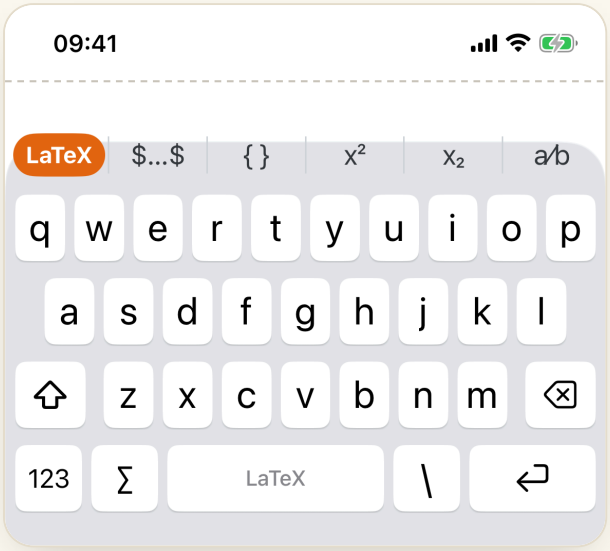
Keys brings the keyboard back.

Your size and background are remembered for next time.

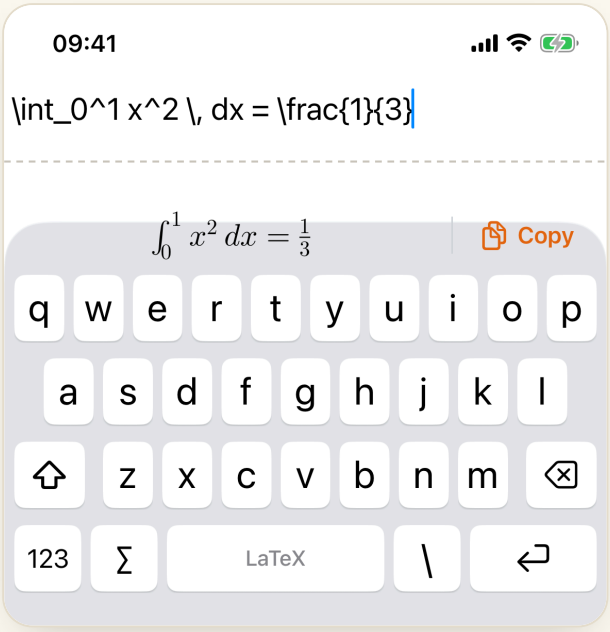


7 LaTeX mode

Writing a LaTeX document, or a long equation? Turn on **LaTeX mode** and nothing turns into symbols at all: `\alpha` stays `\alpha`.



Tap **LaTeX** in the bar. It turns orange, and the space bar says LaTeX.

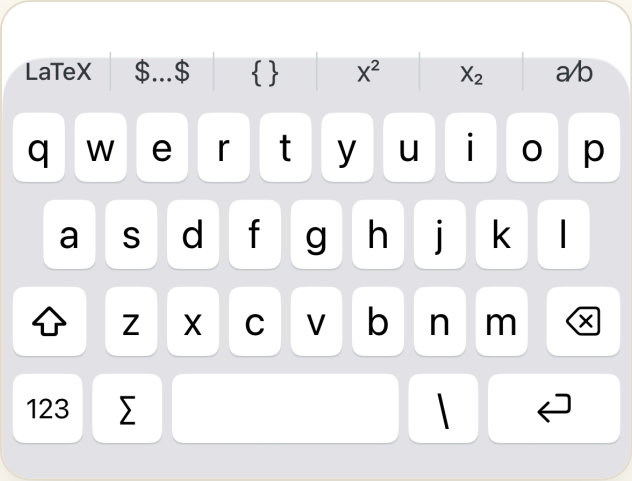


Your LaTeX stays as typed, and the bar still draws it for you to copy.

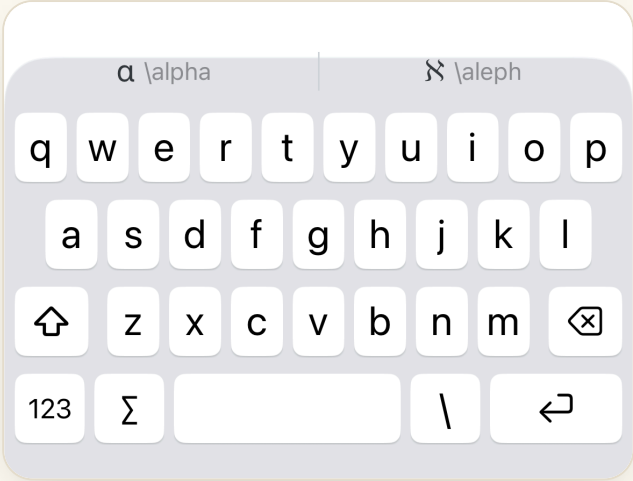
Tap **LaTeX** again to go back to symbols. The keyboard remembers which you chose. In LaTeX mode, the keys on the ∑ page type commands like `\alpha` instead of α.

8 The bar above the keys

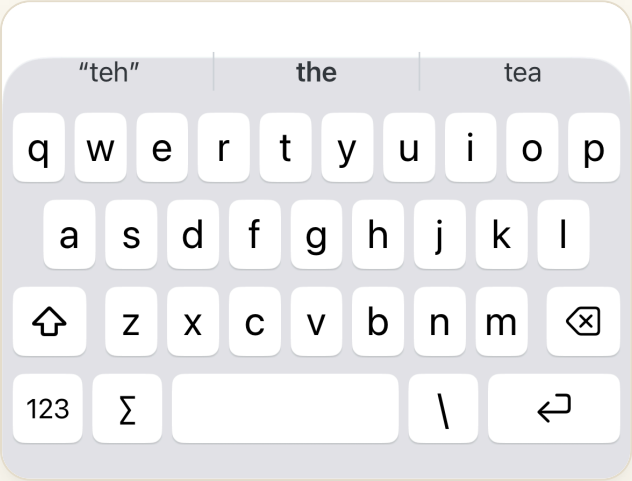
The bar changes with what you are doing.



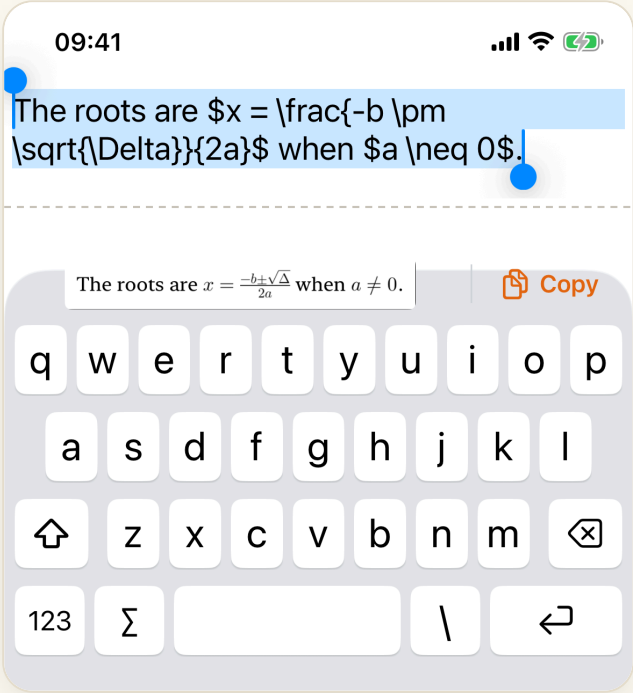
Shortcuts. LaTeX mode, then $…$, braces, x^2, x_2 and a fraction. Templates leave the cursor inside; press space to jump to the next box.



Command suggestions. Start a command and tap the one you mean.



Spelling. Your word, the correction space will make (bold), and another guess. Never inside maths.

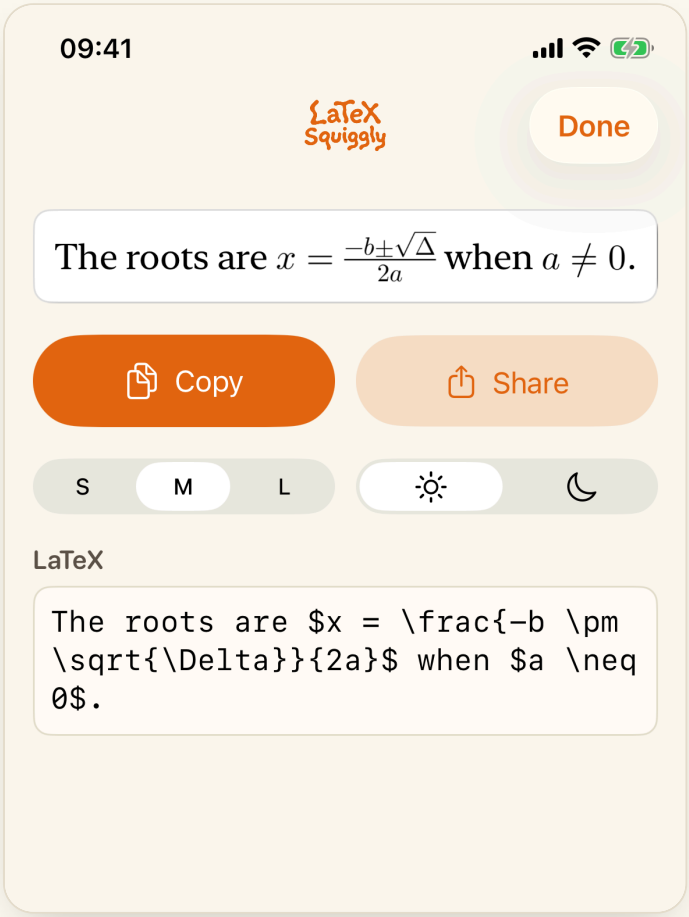


Select and copy. Select any text with maths in it and tap Copy for a picture of exactly that, words included.

9 Render from the share sheet

Got LaTeX somewhere already, in Notes, a web page or a message? Turn it into a picture without retyping it.

1. Select the text.
2. Tap **Share**, then **LaTeX Squiggly**. (The first time, scroll the row of apps to the end, tap **More**, then **Edit**, and add it.)
3. It is drawn straight away. Change the size or background, fix the LaTeX if you like, then **Copy** or **Share**.



Words stay words; the maths between $ signs is typeset.

Plain LaTeX, with no $ signs, becomes one equation.

Text with `$…$` in it keeps its words, with each piece of maths drawn in its line. `$$…$$` goes on a line of its own, in the middle.



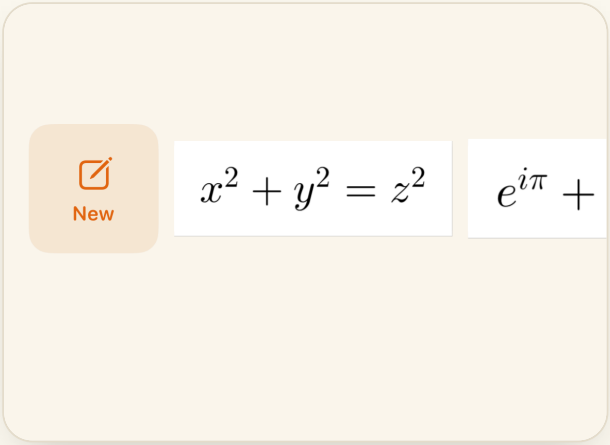
10 Send equations in Messages

In Messages, the picture goes straight into your message: no copy and paste.

1. In a conversation, tap **+**, then **More**, then **LaTeX Squiggly**.
2. Tap **New** and write your LaTeX. It is drawn as you type.
3. Tap **Add to message**, then send.

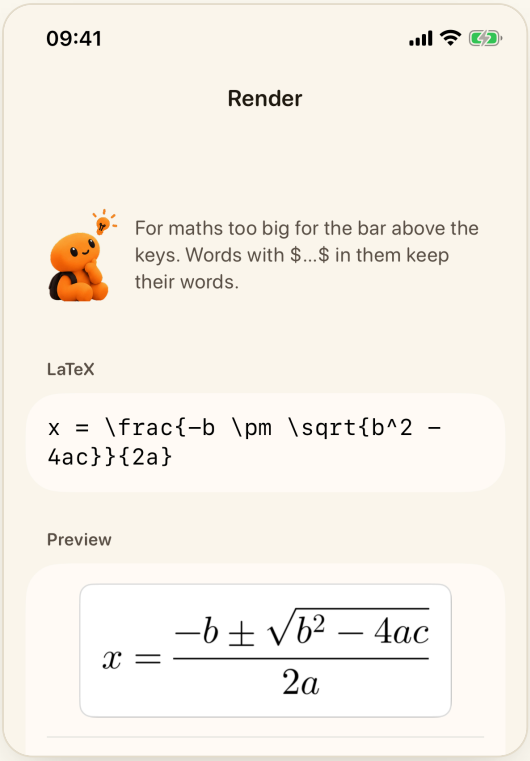


Write, see it drawn, add it to your message.



Equations you sent before wait in the drawer. Tap one to send it again.

11 The app, tips and help



The app's **Render** tab, for longer pieces.

Good to know

- **Nothing appears in the bar?** Check your maths is between $ signs, or that LaTeX mode is on.
- **The bar says "Can't render this"?** It names the command it doesn't know. Check the spelling, or try another way to write it.
- **Copy asks for Full Access?** Turn on Allow Full Access for LaTeX Squiggly in Settings (page 2).
- **The keyboard is missing in a password box?** iPhone always uses its own keyboard there, for safety.
- **No automatic capitals.** LaTeX cares about capitals, so shift only comes on when you tap it. Double-tap for caps lock.
- **Double space** after a word types a full stop.



Happy writing. LaTeX Squiggly is free and open source:
github.com/SuperWalrus01/LaTeX-Squiggly